Requirements Traceability Matrix (RTM)

Functional Requirements (FR) Traceability

Req ID	Requirement	Use Case	Analysis	System Design	Object Design	Implementation (TBF)	Test Case
FR-01	Students register with name, email, password	UC-01: Manage Account	Analysis Class Diagram	HD (System decomposition & Components diagram)	LD (Physical ERD)		TC-01, TC-03
FR-02	Tutors register with name, email, password, bio	UC-01: Manage Account	Analysis Class Diagram	HD (System decomposition & Components diagram)	LD (Physical ERD)		TC-02, TC-03
FR-03	Users log in with email and password	UC-01: Manage Account	Analysis Class Diagram	HD (System decomposition & Components diagram)	LD (Physical ERD)		TC-04, TC-05
FR-04	Tutors add skills from predefined skill list	UC-06: Manage Tutor Profile	Analysis Class Diagram	HD (System decomposition & Components diagram)	LD (Physical ERD)		TC-25, TC-26
FR-05	Tutors upload qualification files (PDF/images)	UC-06: Manage Tutor Profile	Analysis Class Diagram	HD (System decomposition & Components diagram)	LD (Physical ERD)		TC-27, TC-28
FR-06	Students search for tutors by skill name	UC-02: Search for Tutors	Analysis Class Diagram	HD (System decomposition & Components diagram)	LD (Physical ERD)		TC-08, TC-09
FR-07	Tutors set weekly available time slots	UC-06: Manage Tutor Profile	Analysis Class Diagram	HD (System decomposition & Components diagram)	LD (Physical ERD)		TC-29, TC-30
FR-08	Display only tutors available at selected time	UC-02: Search for Tutors	Analysis Class Diagram	HD (System decomposition & Components diagram)	LD (Physical ERD)		TC-10
FR-09	Students send booking request	UC-03: Manage Bookings	Analysis Class Diagram	HD (System decomposition & Components diagram)	LD (Booking Service Class Diagram)		TC-12, TC-13
FR-10	Tutors view incoming booking requests	UC-07: Handle Bookings	Analysis Class Diagram	HD (System decomposition & Components diagram)	LD (Booking Service Class Diagram)		TC-31, TC-32
FR-11	Tutors confirm or reject booking requests	UC-07: Handle Bookings	Analysis Class Diagram	HD (System decomposition & Components diagram)	LD (Booking Service Class Diagram)		TC-33, TC-34, TC-35

Req ID	Requirement	Use Case	Analysis (TBF)	System Design	Object Design	Implementation (TBF)	Test Case
FR-12	Send email notification on confirmation/rejection	UC-03, UC-07	Analysis Class Diagram	HD (System decomposition & Components diagram)	LD (Physical ERD)		TC-16, TC-17
FR-13	Students type natural language request, receive top 3 suggestions	UC-05: View AI Recommendations	Analysis Class Diagram	HD (System decomposition & Components diagram)	LD (AI Recommendations Pipeline)		TC-21, TC-22, TC-24
FR-14	AI considers skills, urgency, rating	UC-05: View AI Recommendations	Analysis Class Diagram	HD (System decomposition & Components diagram)	LD (AI Recommendations Pipeline)		TC-23
FR-15	Students leave rating (1–5 stars) and optional comment	UC-04: Leave Review	Analysis Class Diagram	HD (System decomposition & Components diagram)	LD (Physical ERD)		TC-18, TC-19, TC-20
FR-16	Display average rating and recent reviews on tutor profile	UC-08: View Reviews	Analysis Class Diagram	HD (System decomposition & Components diagram)	LD (Physical ERD)		TC-38, TC-39
FR-17	Students view upcoming and past bookings	UC-03: Manage Bookings	Analysis Class Diagram	HD (System decomposition & Components diagram)	LD (Physical ERD)		TC-14, TC-15
FR-18	Tutors view upcoming sessions and booking history	UC-07: Handle Bookings	Analysis Class Diagram	HD (System decomposition & Components diagram)	LD (Physical ERD)		TC-36, TC-37

Non-Functional Requirements (NFR) Traceability

Req ID	Requirement	Use Case	Analysis (TBF)	System Design	Object Design	Implementation (TBF)	Test Case
NFR-01	Search results return within 3 seconds under 30 concurrent users	—	Analysis Class Diagram	HD (System Architecture)	LD (Physical ERD)		TC-P01
NFR-02	AI recommendations load within 4 seconds under 20 concurrent users	—	Analysis Class Diagram	HD (System Architecture)	LD (AI Recommendations Pipeline)		TC-P02
NFR-03	Passwords hashed using Django's default PBKDF2 algorithm	—	Analysis Class Diagram	HD (System Architecture)	LD (Physical ERD)		TC-S01, TC-S02
NFR-04	All user identifiers use UUID v7; no sensitive data in URL parameters	—	Analysis Class Diagram	HD (System Architecture)	LD (Physical ERD)		TC-S03, TC-S04, TC-S05, TC-S06
NFR-05	System accessible via modern browsers	—	Analysis Class Diagram	HD (System Architecture)	LD (Physical ERD)		TC-C01, TC-C02, TC-C03
NFR-06	UI responsive and usable on mobile devices (375px and above)	—	Analysis Class Diagram	HD (System Architecture)	LD (Physical ERD)		TC-U01, TC-U02, TC-U03, TC-U04
NFR-07	File uploads limited to 5MB, support PDF, JPG, PNG	—	Analysis Class Diagram	HD (System Architecture)	LD (Physical ERD)		TC-U05, TC-U06, TC-U07, TC-U08, TC-U09, TC-U10, TC-U11
NFR-08	System handles up to 50 concurrent users during demo	—	Analysis Class Diagram	HD (System Architecture)	LD (Physical ERD)		TC-SC01
NFR-09	Booking confirmations and email notifications sent within 5 minutes, with 99% delivery success rate	—	Analysis Class Diagram	HD (System Architecture)	LD (Physical ERD)		TC-R01, TC-R02, TC-R03
NFR-10	System meets 99.5% availability during 14-day evaluation period, with max downtime <=2 hours	—	Analysis Class Diagram	HD (System Architecture)	LD (Physical ERD)		TC-A01